

BCH212 Medical Biochemistry

Simulation Examination II

Lectures 10 - 21 | ~150 Questions with Detailed Explanations

Questions marked [PP] are from past papers - prioritise these!

Lecture	Topic	Real Exam	This Sim
Lec 10	DNA & RNA Synthesis	8 Qs	12 Qs
Lec 11	Protein Synthesis & Transport Protein	8 Qs	12 Qs
Lec 12	Liver Function	12 Qs	16 Qs
Lec 13	Thalassemia	6 Qs	10 Qs
Lec 14	Apoptosis	6 Qs	10 Qs
Lec 15	Human Genome & Gene Regulation	8 Qs	12 Qs
Lec 16	Blood & Heme Metabolism	4 Qs	8 Qs
Lec 17	Plasma Protein & Blood Clotting Factor	8 Qs	12 Qs
Lec 18	Nutrition	12 Qs	16 Qs
Lec 19	Genetic Engineering	8 Qs	12 Qs
Lec 20	Free Radical & Antioxidant	4 Qs	8 Qs
Lec 21	Clinical Correlation	10 Qs	14 Qs
TOTAL		94 Qs	152 Qs

[PP] = Past Paper question | Simulate real exam conditions: 94 questions in ~2 hours

LECTURE 10: DNA & RNA SYNTHESIS

Q1. [PP] DNA replication results in two daughter molecules. Which describes their composition?

- A. Each has 2 new strands
- B. Each has 1 new and 1 original strand
- C. Each has 2 original strands
- D. Both are completely new
- E. None of the above

Answer: B

Semi-conservative replication: each daughter molecule keeps one parental strand and gains one newly synthesised strand. Proven by Meselson-Stahl experiment (1958). [PP: SA2 Med34 Q4, Re-exam]

Q2. [PP] Although prokaryotic nucleotide addition is ~20x faster, eukaryotes replicate larger genomes efficiently because:

- A. Less genetic material
- B. Fewer polymerase types
- C. Helicase unwinds more easily
- D. Multiple origins of replication
- E. No exons to replicate

Answer: D

Eukaryotes initiate replication simultaneously at thousands of origins (ori) across all chromosomes. Prokaryotes have only ONE origin per chromosome. This parallelism compensates for slower polymerisation rate. [PP: SA2 Med34 Q3]

Q3. [PP] Okazaki fragments are BEST defined as:

- A. Short RNA primers on the leading strand
- B. Discontinuous DNA segments synthesised on the lagging strand
- C. Long continuous DNA on the leading strand
- D. Telomeric repeat sequences
- E. Protein-DNA complexes at replication fork

Answer: B

The lagging strand is copied 5'→3' in short discontinuous segments (Okazaki fragments) because polymerase can only add 5'→3' but must move in the opposite direction. Each fragment starts with an RNA primer. [PP: SA2 Med34 Q5]

Q4. [PP] Which is TRUE about telomeres?

- A. Located at the chromosome centre
- B. Degraded by telomerase
- C. Do not contain functional protein-coding genes
- D. Lengthen with every replication cycle
- E. All of the above are false

Answer: C

Telomeres are repetitive sequences (TTAGGG x ~2500 in humans) at chromosome ENDS. They protect chromosomes but do NOT code for proteins. Telomerase ELONGATES (not degrades) telomeres. Without telomerase, they shorten with each division. [PP: SA2 Med34 Q1]

Q5. [PP] Base excision repair (BER) is used to repair:

- A. Thymine dimers from UV light
- B. Mismatched bases from replication errors
- C. Single nucleotide damage from alkylation or oxidation
- D. Double-strand breaks
- E. Interstrand crosslinks

Answer: C

BER: removes a single damaged base (alkylated, oxidised, deaminated) using glycosylase. NER removes bulky helix-distorting lesions (UV-induced thymine dimers). MMR corrects replication mismatches. [PP: Key Simulation Q1]

Q6. [PP] Which cancer chemotherapy drug inhibits mammalian Topoisomerase II?

- A. Rifampicin
- B. Amanitin
- C. Chloramphenicol
- D. Adriamycin (Doxorubicin)
- E. Tetracycline

Answer: D

Adriamycin inhibits topoisomerase II, preventing DNA strand religation causing lethal double-strand breaks in rapidly dividing cancer cells. Rifampicin targets bacterial RNA-pol; Amanitin targets eukaryotic RNA-pol II. [PP: Key Simulation Q3]

Q7. [PP] Xeroderma pigmentosum (XP) results from deficiency in which DNA repair pathway?

- A. Base excision repair
- B. Mismatch repair
- C. Nucleotide excision repair
- D. Homologous recombination
- E. Non-homologous end joining

Answer: C

XP: autosomal recessive. NER normally removes bulky UV-induced thymine dimers (TT). Without NER, UV damage accumulates -> multiple skin cancers at young age. Quinolones target bacterial topoisomerase (not NER). [PP: FA2 Sec11 Q3]

Q8. [PP] Which correctly distinguishes replication from transcription?

- A. Transcription requires a primer; replication does not
- B. Replication requires a primer; transcription does not
- C. Both use the same enzyme
- D. Transcription uses dNTPs as substrate
- E. Both replicate the entire DNA molecule

Answer: B

DNA polymerase CANNOT start de novo - it needs an RNA primer (made by primase). RNA polymerase CAN initiate transcription without a primer. Transcription uses NTPs (ribonucleotides); replication uses dNTPs (deoxyribonucleotides). [PP: Key Simulation Q4]

Q9. [PP] What is the coding strand DNA sequence for mRNA 5'-CCAAUUGGCGCGUUCCCGUACGCC-3'?

- A. 5' CCAAUUGGCGCGUUCCCGUACGCC 3'
- B. 5' GGCGTACGGGAACGCGCCAATTG 3'
- C. 5' CCAATTGGCGCGTTCCCGTACGCC 3'
- D. 3' CCAATTGGCGCGAACCCGTACGCC 5'
- E. 5' GGUUAACCGCGCAAGGGCAUGCGG 3'

Answer: C

The coding (sense) strand has the same sequence as mRNA but with T instead of U. Replace all U with T: 5'-CCAATTGGCGCGTTCCCGTACGCC-3'. The antisense (template) strand is read 3'→5' by RNA polymerase. [PP: SA2 Med34 Q2, Re-exam Q9]

Q10. Which enzyme introduces negative supercoiling in bacterial DNA and is the target of quinolone antibiotics?

- A. DNA ligase
- B. DNA helicase
- C. DNA gyrase (Topoisomerase II)
- D. DNA polymerase I
- E. DNA primase

Answer: C

DNA gyrase (a type II topoisomerase) introduces negative supercoils ahead of the replication fork to relieve tension. Quinolones (ciprofloxacin, nalidixic acid) inhibit gyrase, blocking bacterial DNA replication. [PP: FA2 Sec11 Q1]

Q11. What type of mutation adds or removes 1-2 nucleotides, changing the reading frame?

- A. Silent mutation
- B. Missense mutation
- C. Nonsense mutation
- D. Frameshift mutation (insertion/deletion)
- E. Conservative mutation

Answer: D

Insertion or deletion of 1 or 2 nucleotides (not multiples of 3) shifts the reading frame for all downstream codons. This typically produces a severely altered or truncated protein. Adding/deleting 3 nt adds/removes one amino acid without frameshifting. [PP: FA2 Sec11 Q, Re-exam Q4]

Q12. mRNA vaccine for SARS-CoV-2 works by which mechanism?

- A. Deleting the virus target gene
- B. Being translated by ribosomes to express spike protein stimulating immunity
- C. Binding to DNA for transcription
- D. Regulating host DNA directly
- E. Reverse transcribing into host genome

Answer: B

mRNA vaccines (Pfizer-BioNTech, Moderna) deliver lipid nanoparticle-encapsulated mRNA encoding the spike protein. Host ribosomes TRANSLATE the mRNA → spike protein → immune response. The mRNA does NOT enter the nucleus or alter DNA. It is eventually degraded. [PP: SA2 MED35 Q8]

LECTURE 11: PROTEIN SYNTHESIS & TRANSPORT PROTEIN

Q13. [PP] Which release factor recognises the UGA stop codon during bacterial translation?

- A. RF1
- B. RF2
- C. RF3
- D. EF-G
- E. EF-Tu

Answer: B

Bacterial release factors: RF1 recognises UAA and UAG; RF2 recognises UAA and UGA. RF3 is a GTPase that assists recycling of RF1/RF2. In eukaryotes, a single release factor (eRF1) recognises all three stop codons.

[PP: SA2 Med34 Q2, Re-exam Q18]

Q14. [PP] Which biomolecule charges tRNA with the correct amino acid?

- A. 16S rRNA
- B. 23S rRNA
- C. 28S rRNA
- D. Aminoacyl-tRNA synthase (incorrect name)
- E. Aminoacyl-tRNA synthetase

Answer: E

Aminoacyl-tRNA synthetases (one per amino acid, 20 total) use ATP to attach the correct amino acid to its cognate tRNA. Step 1: aa + ATP → aminoacyl-AMP + PP_i. Step 2: aminoacyl-AMP + tRNA → aminoacyl-tRNA + AMP. [PP: FA2 Sec11 Q5]

Q15. [PP] Ricin (toxin from castor beans) causes protein synthesis failure by:

- A. Blocking aminoacyl-tRNA binding to A site
- B. Depurinating and cleaving 28S rRNA of the large ribosomal subunit
- C. Inhibiting peptidyl transferase
- D. Blocking translocation
- E. Inhibiting initiation factor eIF2

Answer: B

Ricin is an N-glycosidase that removes a specific adenine from 28S rRNA in the large (60S) ribosomal subunit. This permanently inactivates the ribosome. Even one molecule of ricin can kill a cell. [PP: SA2 Med34 Q1, Re-exam Q17]

Q16. [PP] The initiator aminoacyl-tRNA binds which site during translation initiation?

- A. E site (exit)
- B. A site (aminoacyl)
- C. P site (peptidyl)
- D. Decoding centre
- E. Exit tunnel

Answer: C

The initiator Met-tRNA (eukaryotes) or fMet-tRNA (prokaryotes) DIRECTLY occupies the P site during initiation. This is unique to the initiator tRNA. All subsequent aminoacyl-tRNAs enter the A site during elongation. [PP: SA2 Med34 Q8, Re-exam Q21]

Q17. [PP] Which antibiotic blocks translation in BOTH prokaryotes AND eukaryotes?

- A. Chloramphenicol
- B. Tetracycline
- C. Puromycin
- D. Streptomycin
- E. Erythromycin

Answer: C

Puromycin is an aminoacyl-tRNA analogue. It inserts into the A site and forms a peptide bond with the growing chain, causing premature chain release. It acts on BOTH 70S and 80S ribosomes. [PP: Key Simulation Q6, Re-exam Q25]

Q18. [PP] Which molecule guides the ribosome-mRNA-nascent protein complex to the ER for proteins destined for secretion?

- A. RF1
- B. RF2
- C. Signal Recognition Particle (SRP)
- D. EF-2
- E. IF2

Answer: C

Proteins for secretion carry a hydrophobic N-terminal signal peptide. SRP recognises this signal co-translationally and escorts the entire translation complex to the SRP receptor on the ER membrane. Translation resumes into the ER lumen. Signal peptide is then cleaved. [PP: SA2 MED35 Q18]

Q19. [PP] Tetracycline's antibacterial mechanism is:

- A. Inhibiting peptidyl transferase in 50S subunit
- B. Blocking aminoacyl-tRNA binding to the 30S A site
- C. Causing mRNA misreading
- D. Inhibiting translocation by EF-G
- E. Cleaving 16S rRNA

Answer: B

Tetracycline chelates Mg^{2+} and binds the 30S ribosomal subunit near the A site, blocking aminoacyl-tRNA from binding. This halts elongation. It is bacteriostatic (not bactericidal). [PP: Med36 DNA&RNA; section, Re-exam Q20]

Q20. [PP] Which protein does NOT require a signal sequence?

- A. Insulin
- B. Growth hormone
- C. Cytochrome P450 (ER)
- D. GLUT4
- E. Pyruvate kinase

Answer: E

Pyruvate kinase is a CYTOSOLIC enzyme functioning in glycolysis. Cytosolic proteins are translated on free ribosomes and remain in the cytoplasm — no signal sequence needed. Insulin, growth hormone are secreted proteins; GLUT4 is a membrane transporter; Cytochrome P450 is ER-resident. [PP: SA2 Med34 Q7, Re-exam Q20]

Q21. [PP] Given mRNA: 5'-AUG UGG CAU CGU UAA-3'. How many amino acids are in the peptide?

- A. 3
- B. 4
- C. 5
- D. 6
- E. 7

Answer: B

AUG (start/Met) - UGG (Trp) - CAU (His) - CGU (Arg) - UAA (STOP). The stop codon (UAA) does not code for an amino acid. So 4 amino acids: Met-Trp-His-Arg. [PP: FA2 Sec11 Q4, FA2 Med33-34 Q4]

Q22. Chloramphenicol inhibits which enzymatic activity?

- A. Aminoacyl-tRNA synthetase
- B. Peptidyl transferase (in 50S subunit)
- C. Translocation by EF-G
- D. mRNA decoding
- E. Ribosome assembly

Answer: B

Chloramphenicol binds the 50S ribosomal subunit and inhibits peptidyl transferase — the enzymatic activity of 23S rRNA that forms peptide bonds. It is used for serious infections but can cause aplastic anaemia. Also inhibits mitochondrial ribosomes (70S) explaining some toxicity. [PP: Med36, Re-exam Q21]

Q23. In eukaryotes, what is the size of the complete ribosome?

- A. 50S
- B. 60S
- C. 70S
- D. 80S
- E. 100S

Answer: D

Eukaryotic ribosome: 80S = 40S (small subunit) + 60S (large subunit). Prokaryotic ribosome: 70S = 30S + 50S. Mitochondrial ribosomes are 55S (similar to prokaryotic). S = Svedberg units (sedimentation coefficient). [PP: Key Simulation Lec11 Q1]

Q24. Which elongation factor escorts aminoacyl-tRNA into the ribosome A site during elongation?

- A. IF-Ts-GTP
- B. EF-Ts-ATP
- C. EF-Tu-GTP (EF1A in eukaryotes)
- D. EF-G
- E. RF2

Answer: C

EF-Tu (prokaryotes) / eEF1A (eukaryotes) binds aminoacyl-tRNA + GTP and delivers it to the A site. GTP hydrolysis triggers conformational change and EF-Tu-GDP release. EF-Ts regenerates EF-Tu-GTP. EF-G (eEF2) mediates translocation. [PP: SA2 Med34 Q4, Re-exam Q19]

LECTURE 12: LIVER FUNCTION

Q25. [PP] What is the rate-limiting enzyme of bile acid synthesis?

- A. HMG-CoA reductase
- B. Cholesterol 7- α -hydroxylase (CYP7A1)
- C. Sterol 27-hydroxylase
- D. UDP-glucuronyl transferase
- E. Heme oxygenase

Answer: B

CYP7A1 catalyses the first and committed step converting cholesterol to bile acids. Found ONLY in hepatocytes. Upregulated by cholesterol; downregulated by bile salts (feedback). Statins target HMG-CoA reductase (cholesterol synthesis, different pathway). [PP: SA2 MED35 Q66, SA2 Med34 Q4 Lec15]

Q26. [PP] Which molecule conjugates with bilirubin to make it water-soluble for bile excretion?

- A. Fatty acid
- B. Acetyl-CoA
- C. Glucuronic acid
- D. Sulphate
- E. Albumin

Answer: C

UDP-glucuronyl transferase (UGT1A1) adds glucuronic acid to unconjugated bilirubin -> bilirubin diglucuronide (conjugated, water-soluble). Conjugated bilirubin is excreted into bile. Deficiency: Crigler-Najjar (severe) or Gilbert's syndrome (mild). [PP: FA2 Med33-34 Q12, SA2 MED35 Q88, Re-exam Q3]

Q27. [PP] In thalassemia, the predominant form of elevated serum bilirubin is:

- A. Conjugated bilirubin
- B. Unconjugated bilirubin
- C. Bilirubin bound to apolipoprotein
- D. Bilirubin bound to bile acids
- E. Bilirubin bound to globulin

Answer: B

Thalassemia causes haemolysis -> excess haem -> UNCONJUGATED bilirubin overwhelms liver conjugation capacity. Pre-hepatic (haemolytic) jaundice = elevated UNCONJUGATED bilirubin. Skin/sclera yellow but urine not dark (unconjugated cannot be filtered by kidney). [PP: FA2 Sec11 Q10]

Q28. [PP] Which enzyme digests fibrin clots (fibrinolysis)?

- A. Fibrinogen
- B. Plasmin
- C. Dicoumarol
- D. Warfarin
- E. Thrombin

Answer: B

Plasmin is the serine protease that breaks down fibrin. Plasminogen -> plasmin is catalysed by t-PA (tissue plasminogen activator). Thrombin converts fibrinogen to fibrin (forming clots). Warfarin/dicoumarol prevent clot formation. Streptokinase/urokinase activate plasminogen. [PP: FA2 Sec11 Q17]

Q29. [PP] A 50-year-old with jaundice and bile duct gallstones most likely shows elevated:

- A. Serum albumin
- B. Gamma-glutamyl transferase (GGT) and Alkaline phosphatase (ALP)
- C. Alpha-fetoprotein
- D. Glucuronic acid
- E. AST and ALT

Answer: B

Post-hepatic (obstructive/cholestatic) jaundice = bile duct obstruction. GGT and ALP are enzymes from biliary epithelium cells that leak when bile backs up. AST/ALT = hepatocellular injury markers. AFP = liver cancer marker. Albumin decreases in chronic disease. [PP: SA2 MED35 Q63]

Q30. [PP] Disulfiram (Antabuse) treats alcoholism by inhibiting:

- A. Alcohol dehydrogenase (ADH)
- B. Aldehyde dehydrogenase (ALDH)
- C. CYP2E1 (MEOS)
- D. Acetyl-CoA synthetase
- E. Pyruvate dehydrogenase

Answer: B

Ethanol -> (ADH) -> acetaldehyde -> (ALDH, mitochondria) -> acetate. Disulfiram blocks ALDH -> acetaldehyde accumulates -> flushing, tachycardia, nausea, vomiting, hypotension. This aversive reaction deters drinking. [PP: SA2 MED35, Med36 Lec12 Q5, Re-exam Q19]

Q31. [PP] Which is the biological effect of excess NADH generated by ethanol metabolism?

- A. Increased gluconeogenesis
- B. Activated TCA cycle
- C. Increased fatty acid synthesis (fatty liver)
- D. Decreased lactic acid production
- E. Increased beta-oxidation of fatty acids

Answer: C

Ethanol -> large amounts of NADH. High NADH/NAD⁺: inhibits gluconeogenesis, inhibits TCA cycle, inhibits fatty acid beta-oxidation, promotes fatty acid SYNTHESIS, promotes lactate (lactic acidosis). Net: triglyceride accumulation -> fatty liver -> alcoholic hepatitis -> cirrhosis. [PP: SA2 Med34 Q8, FA2 Sec11 Q11]

Q32. [PP] Which is a marker for LIVER CANCER in adults?

- A. Albumin
- B. Prothrombin
- C. Alpha-fetoprotein (AFP)
- D. Haptoglobin
- E. C-reactive protein

Answer: C

AFP is normally produced by fetal liver/yolk sac; very low in adults (<10 ng/mL). Markedly elevated AFP in adults suggests hepatocellular carcinoma (HCC), testicular/ovarian germ cell tumors, or metastatic liver disease. Also elevated in pregnancy. [PP: SA2 Med34 Q6, Re-exam Q10]

Q33. [PP] Which is classified as a SECONDARY bile acid?

- A. Cholic acid
- B. Chenodeoxycholic acid
- C. Lithocholic acid
- D. Glycocholic acid
- E. Taurocholic acid

Answer: C

PRIMARY bile acids (liver): cholic acid and chenodeoxycholic acid. SECONDARY bile acids (intestinal bacteria remove a hydroxyl group): deoxycholic acid (from cholic) and LITHOCHOLIC acid (from chenodeoxycholic). Glycocholic/taurocholic are conjugated primary bile acids. [PP: SA2 Med34 Q7, FA2 Sec11 Q12]

Q34. [PP] Phase II biotransformation in the liver involves:

- A. Oxidation by CYP450
- B. Reduction reactions
- C. Conjugation (glucuronidation, glutathione conjugation, sulphation)
- D. Hydrolysis
- E. Hydroxylation

Answer: C

Phase I (CYP450): oxidation/reduction/hydrolysis - introduces reactive group. Phase II (transferases): conjugation with glucuronic acid, glutathione, sulphate, acetate, amino acids -> makes compound more hydrophilic for excretion. Key enzyme: Glutathione S-Transferase (GST), UGT. [PP: SA2 Med34 Q3, Med36 Lec12 Q7]

Q35. [PP] Crigler-Najjar syndrome Type I is due to complete deficiency of:

- A. Heme oxygenase
- B. Biliverdin reductase
- C. UDP-glucuronyl transferase (UGT1A1)
- D. Cholesterol 7- α -hydroxylase
- E. ALT

Answer: C

Without UGT1A1, unconjugated bilirubin cannot be conjugated -> severe hyperbilirubinemia from birth -> kernicterus (brain damage) -> fatal without liver transplant. Type II = partial UGT1A1 deficiency (responds to phenobarbital). Gilbert's = mild reduction (~10-30%), usually benign. [PP: Med36 Lec12 Q6]

Q36. [PP] Which enzyme is responsible for MEOS (Microsomal Ethanol Oxidising System)?

- A. Alcohol dehydrogenase
- B. Aldehyde dehydrogenase
- C. CYP2E1
- D. CYP3A4
- E. Xanthine oxidase

Answer: C

MEOS = CYP2E1 (cytochrome P450 2E1) in the smooth ER (microsome). Induced at HIGH ethanol concentrations and with chronic use. Uses NADPH and O₂, generates reactive oxygen species (ROS) -> oxidative stress -> liver damage. Explains why chronic drinkers metabolise alcohol faster. [PP: Med36 Lec12 Q5]

Q37. Which test BEST reflects hepatic SYNTHETIC function?

- A. ALT (alanine aminotransferase)
- B. GGT (gamma-glutamyl transferase)
- C. Alkaline phosphatase (ALP)
- D. Serum albumin and prothrombin time (PT)
- E. Total bilirubin

Answer: D

Synthetic function = what the liver MAKES. Albumin (exclusively by hepatocytes, 12g/day) and clotting factors (PT reflects factors II, VII, IX, X). ALT/AST = hepatocellular INJURY. GGT/ALP = biliary markers. Bilirubin = excretory function. In cirrhosis: low albumin + prolonged PT = poor synthetic function.

Q38. The enterohepatic circulation refers to:

- A. Blood flow from intestine to liver via portal vein
- B. Recycling of bile acids between liver, bile, intestine, and back to liver via portal vein
- C. Transport of glucose from intestine to liver
- D. Lymphatic drainage of the liver
- E. Enterocyte renewal cycle

Answer: B

95% of bile acids secreted into intestine are reabsorbed at the terminal ileum, returned to liver via portal vein, re-conjugated and re-secreted. Only 5% are newly synthesised. This recycling conserves bile acids (pool recycled 6-10x/day). Cholestyramine binds bile acids in gut, preventing reabsorption -> increases cholesterol conversion to new bile acids -> lowers serum cholesterol.

Q39. Heme oxygenase catalyses what reaction in bilirubin metabolism?

- A. Converts biliverdin -> bilirubin
- B. Converts haemoglobin -> haem
- C. Opens the porphyrin ring of haem -> biliverdin (rate-limiting step of bilirubin production)
- D. Conjugates bilirubin with glucuronic acid
- E. Converts bilirubin -> urobilinogen

Answer: C

Heme oxygenase (in reticuloendothelial system): haem + O₂ + NADPH -> biliverdin (green) + CO + Fe²⁺. This is the RATE-LIMITING step. Biliverdin reductase then converts biliverdin -> bilirubin (yellow). Iron is recycled; CO is exhaled. 85% of bilirubin comes from senescent RBC haemoglobin.

Q40. In which type of jaundice is urobilinogen absent from urine and stool is pale (clay-coloured)?

- A. Pre-hepatic jaundice (haemolytic)
- B. Hepatic jaundice (hepatocellular)
- C. Post-hepatic jaundice (obstructive/cholestatic)
- D. Neonatal physiological jaundice
- E. Gilbert's syndrome

Answer: C

Post-hepatic (obstructive) jaundice: bile duct blocked -> conjugated bilirubin cannot reach intestine -> NO urobilinogen formed -> pale/clay-coloured stool. Conjugated bilirubin backs up into blood -> dark urine. Pre-hepatic: increased urobilinogen in urine and stool (dark). Hepatic: mixed picture.

LECTURE 13: THALASSEMIA

Q41. [PP] What is the genotype of Hb Bart's hydrops fetalis?

- A. $\alpha\alpha\alpha\alpha$ (3-gene deletion)
- B. $\alpha\alpha\alpha\alpha$ (homozygous α -thal2)
- C. $\alpha\alpha\alpha\alpha$ (all 4 α genes deleted)
- D. $\alpha\alpha\alpha\alpha$ (alpha-thal1 trait)
- E. $\alpha\alpha\alpha\alpha$

Answer: C

Hb Bart's = ALL 4 α -globin genes absent ($\alpha\alpha\alpha\alpha$). Without α chains, γ chains form tetramers (γ_4 = Hb Bart's) with extremely high O₂ affinity $\rightarrow$ severe tissue hypoxia. Fatal in utero or shortly after birth (hydrops fetalis). [PP: SA2 MED35, SA2 Med34, Med36, Re-exam]

Q42. [PP] Which genotype results in Hb H disease?

- A. $\alpha\alpha\alpha\alpha$ (all 4 deleted)
- B. $\alpha\alpha\alpha\alpha$ (3 of 4 α genes deleted)
- C. $\alpha\alpha\alpha\alpha$ (2 deleted)
- D. $\alpha\alpha\alpha\alpha$ (2 deleted on same chromosome)
- E. $\alpha\alpha\alpha\alpha$ (normal)

Answer: B

Hb H disease = 3 of 4 α genes non-functional ($\alpha\alpha\alpha\alpha$). Only ONE functional α gene remains. Excess β chains form β_4 tetramers = Hb H. Causes moderate-severe haemolytic anaemia. [PP: SA2 Med34, Key Simulation Q4]

Q43. [PP] Hydroxyurea pharmacologically stimulates HbF production in beta-thalassemia because:

- A. It chelates excess iron
- B. It activates γ -globin gene expression increasing HbF ($\alpha_2\gamma_2$)
- C. It inhibits α chain synthesis
- D. It directly replaces β chains
- E. It suppresses erythropoiesis

Answer: B

Hydroxyurea (and 5-azacytidine, butyrate, erythropoietin) reactivates γ -globin gene expression $\rightarrow$ increased HbF ($\alpha_2\gamma_2$) production. HbF compensates for deficient adult HbA in beta-thalassemia and sickle cell disease. [PP: SA2 Med34, Key Simulation Q2&Q5;]

Q44. [PP] Father is beta-thalassemia trait, mother is homozygous HbE. Probability their child has thalassemia disease?

- A. 0%
- B. 25%
- C. 50%
- D. 75%
- E. 100%

Answer: C

Father: $\beta\beta\beta\beta$ carrier. Mother: $\beta\beta\beta\beta$. Child inherits: from father 50% $\beta\beta\beta\beta$ (or 50% $\beta\beta\beta\beta$); from mother always $\beta\beta\beta\beta$. $\beta\beta\beta\beta/\beta\beta\beta\beta$ = β -thal/HbE disease (50%). $\beta\beta\beta\beta/\beta\beta\beta\beta$ = HbE trait (50%). Result: 50% thalassemia disease. [PP: SA2 Med34 Q2, Med36 Q5]

Q45. [PP] Hemoglobin Constant Spring (HbCS) is caused by:

- A. Deletion of both alpha genes on one chromosome
- B. Missense mutation in the beta-globin gene codon 26
- C. Point mutation at alpha-globin stop codon 142: UAA -> CAA
- D. Frameshift mutation in exon 2 of alpha-globin
- E. Splice site mutation causing exon skipping

Answer: C

HbCS: normal alpha-globin stop codon (UAA at codon 142) mutates to CAA (Glutamine). Translation CONTINUES past normal stop -> abnormally LONG alpha chain (172 aa instead of 141). Non-deletion alpha-thalassemia mutation. Found in 1-8% of Thais. [PP: SA2 MED35, Key Simulation Q4]

Q46. [PP] Which hemoglobin is predominant in FETAL blood?

- A. HbA (alpha2-beta2)
- B. HbF (alpha2-gamma2)
- C. HbA2 (alpha2-delta2)
- D. Hb Gower 1 (zeta2-epsilon2)
- E. Hb Portland (zeta2-gamma2)

Answer: B

HbF (alpha2gamma2) dominates from 2nd trimester to birth (~97% at birth). Higher O2 affinity than HbA facilitates O2 transfer from maternal to fetal circulation. Rapidly replaced by HbA by 6-12 months of age. In adult: HbA 97%, HbA2 2.5%, HbF 0.5%. [PP: SA2 Med34, SA2 MED35 Q50/Q80, Med36]

Q47. [PP] Which hemoglobin is found in the EMBRYO (earliest development)?

- A. alpha2-beta2
- B. alpha2-gamma2
- C. alpha2-delta2
- D. alpha2-zeta2
- E. zeta2-epsilon2

Answer: E

Embryonic haemoglobins (yolk sac, first 8 weeks): Hb Gower 1 (zeta2-epsilon2), Hb Gower 2 (alpha2-epsilon2), Hb Portland (zeta2-gamma2). Zeta = embryonic alpha-like; epsilon = embryonic beta-like. [PP: SA2 Med34 Q1, Key Simulation Q6, Re-exam Q25]

Q48. [PP] Repeated blood transfusions in severe thalassemia most commonly leads to:

- A. Hyperkalemia
- B. Iron overload (hemosiderosis) damaging heart, liver, endocrine glands
- C. Folate deficiency
- D. Elevated HbF
- E. Thrombocytopenia

Answer: B

Each unit of blood contains ~200-250 mg iron. Body cannot excrete excess iron -> accumulates in heart (cardiomyopathy), liver (cirrhosis), pancreas (diabetes), pituitary (hypogonadism). Iron chelation therapy (deferrioxamine/deferiprone/deferasirox) is essential for transfused patients. [PP: Med36, Lec13]

Q49. In beta-zero thalassemia/HbE (the most common severe thalassemia in Thailand), the beta-E mutation involves:

- A. Complete deletion of the beta-globin gene
- B. Codon 26 GAG->AAG (Glu->Lys) creating a cryptic splice site
- C. Point mutation at codon 142 (stop codon)
- D. Insertion of TTCT at codon 41/42
- E. Promoter deletion reducing beta-globin expression

Answer: B

HbE mutation: codon 26 GAG (Glu) -> AAG (Lys) in the beta-globin gene. This missense mutation ALSO creates a cryptic splice site in exon 1, causing some abnormal splicing -> reduced normal beta-globin mRNA. When combined with beta0-thalassemia, causes severe disease requiring transfusions. Common in SE Asia.

Q50. Alpha-thalassemia trait (carrier) with 2 deleted alpha genes can result from which genotypes?

- A. --/alpha-alpha (cis deletion, alpha-thal1) only
- B. -alpha/-alpha (trans deletion, alpha-thal2) only
- C. Either --/alpha-alpha or -alpha/-alpha
- D. --/-- only
- E. alpha-CS-alpha/alpha-alpha only

Answer: C

Alpha-thalassemia trait (2 genes deleted): CIS: both deletions on same chromosome (--/alpha-alpha = alpha-thal1 heterozygote); TRANS: one deletion on each chromosome (-alpha/-alpha = alpha-thal2 homozygote). Both have 2 functional alpha genes -> mild/no anaemia. BUT RISK DIFFERS for offspring: cis carriers can have Hb Bart's babies if both parents carry --.

LECTURE 14: APOPTOSIS

Q51. [PP] Which feature is characteristic of APOPTOSIS but NOT necrosis?

- A. Cell swelling
- B. Nuclear fragmentation and chromatin condensation
- C. Membrane rupture releasing cell contents
- D. Significant inflammation
- E. Random DNA degradation

Answer: B

Apoptosis (programmed, energy-dependent): cell shrinkage, chromatin condensation, nuclear fragmentation, membrane blebbing -> apoptotic bodies phagocytosed WITHOUT inflammation. Necrosis (passive): cell swelling, membrane rupture, contents released -> inflammation. [PP: SA2 Med34 Q1, Med36 Q4, Re-exam Q6]

Q52. [PP] Which component is NOT part of the apoptosome?

- A. Cytochrome C
- B. APAF-1
- C. dATP
- D. Pro-caspase 9
- E. Pro-caspase 10

Answer: E

Apoptosome (intrinsic pathway) = Cytochrome C + APAF-1 + dATP -> forms a wheel-like heptamer recruiting and activating pro-caspase 9 (initiator). Pro-caspase 10 is involved in the EXTRINSIC pathway (DISC complex), not in the apoptosome. [PP: SA2 Med34 Q4, Key Simulation Q4, Re-exam Q8]

Q53. [PP] Which MAPK pathway kinase promotes apoptosis?

- A. ERK1/2 (survival/proliferation)
- B. p38 MAPK (stress-activated, pro-apoptotic)
- C. AKT/PKB (survival)
- D. PDK1 (survival)
- E. mTOR (growth)

Answer: B

MAPK family: ERK1/2 = survival and proliferation (anti-apoptotic in most contexts). p38 MAPK and JNK = stress-activated by UV, cytokines, oxidative stress -> pro-apoptotic. AKT is a major survival kinase (phosphorylates BAD -> inactivates it). [PP: SA2 Med34 Q3, Med36 Q5, Re-exam Q3]

Q54. [PP] Which protein in the Bcl-2 family is PRO-apoptotic?

- A. Bcl-2
- B. BAX
- C. Mcl-1
- D. Bcl-w
- E. Bcl-xL

Answer: B

Anti-apoptotic Bcl-2 family (suppress death): Bcl-2, Bcl-xL, Mcl-1, Bcl-w, Bcl-B. Pro-apoptotic: Multidomain: BAX, BAK (form pores in mitochondrial membrane -> Cyt C release). BH3-only: BAD, BID, PUMA, NOXA (neutralise anti-apoptotic proteins). [PP: SA2 Med34 Q7, Key Simulation Q7, Re-exam Q2]

Q55. [PP] Hoechst 33342 staining is used in apoptosis detection to observe:

- A. Mitochondrial membrane potential
- B. Phosphatidylserine externalisation
- C. Nuclear morphology (chromatin condensation and fragmentation)
- D. DNA content for cell cycle analysis
- E. Protein expression by Western blot

Answer: C

Hoechst 33342 is a cell-permeable DNA dye binding AT-rich minor groove regions. Under fluorescence microscopy, apoptotic cells show condensed, bright, fragmented nuclei vs. normal diffuse staining. Annexin V detects PS; PI (flow cytometry) measures DNA content; JC-1 measures mitochondrial potential. [PP: SA2 Med34 Q5, SA2 MED35, Re-exam Q9]

Q56. [PP] In flow cytometric cell cycle analysis, apoptotic cells appear in which population?

- A. G0/G1 peak
- B. Sub-G1 peak (left of G1)
- C. S phase
- D. G2/M peak
- E. Hyperdiploid peak

Answer: B

Apoptotic DNA fragmentation: DNA is cleaved into nucleosome-sized fragments by endonucleases. These small fragments can wash out during staining -> less DNA than normal G1 cells -> appear as 'sub-G1' population to the LEFT of the G1 peak. Used as a quantitative measure of apoptosis. [PP: SA2 Med34 Q6, Re-exam Q10]

Q57. [PP] The KEY initiator caspase of the EXTRINSIC apoptosis pathway is:

- A. Caspase 3 (executioner)
- B. Caspase 6 (executioner)
- C. Caspase 7 (executioner)
- D. Caspase 8 (initiator, extrinsic)
- E. Caspase 9 (initiator, intrinsic)

Answer: D

Extrinsic pathway: death receptor (Fas, TRAIL-R, TNFR) ligation -> DISC (Death-Inducing Signalling Complex) forms -> recruits and activates pro-caspase 8. Active caspase 8 cleaves executioner caspases (3,7) directly or cleaves BID (linking to intrinsic). Caspase 9 = intrinsic pathway initiator. [PP: SA2 Med34, Key Simulation Q3]

Q58. [PP] Annexin V fluorescence in flow cytometry detects:

- A. DNA fragmentation
- B. Phosphatidylserine (PS) externalisation on outer membrane leaflet
- C. Mitochondrial membrane potential loss
- D. Cytochrome C release
- E. Caspase 3 activation

Answer: B

In healthy cells, PS is on the INNER leaflet of plasma membrane. During EARLY apoptosis, flippase activity stops and scramblase activates -> PS flips to outer surface. Annexin V (Ca²⁺-dependent) binds exposed PS. Combined with PI (membrane integrity marker): Annexin+/PI- = early apoptosis; Annexin+/PI+ = late apoptosis/necrosis. [PP: Lec14 slide, Re-exam]

Q59. PARP (Poly ADP-ribose polymerase) cleavage during apoptosis results in:

- A. Activation of DNA repair, promoting cell survival
- B. Inactivation of PARP, impairing DNA repair -> promoting cell death
- C. Release of cytochrome C from mitochondria
- D. Activation of initiator caspases
- E. Formation of apoptotic bodies

Answer: B

Active PARP detects and repairs DNA strand breaks (cell survival). During apoptosis, caspase 3 and 7 CLEAVE PARP (116 kDa) into an 89 kDa fragment and a 27 kDa fragment. Cleavage INACTIVATES PARP's DNA repair function, committing the cell to death. PARP cleavage is a hallmark of caspase activation and apoptosis (detected by Western blot). [PP: Key Simulation Q, Lec14]

Q60. Bcl-2 anti-apoptotic proteins prevent cytochrome C release by:

- A. Directly inhibiting caspase 3 and 7
- B. Preventing BAX/BAK from forming pores in the outer mitochondrial membrane
- C. Blocking death receptor signalling
- D. Degrading APAF-1
- E. Exporting cytochrome C from the mitochondria

Answer: B

Bcl-2 family anti-apoptotic proteins (Bcl-2, Bcl-xL, Mcl-1) BIND pro-apoptotic BAX and BAK, preventing them from oligomerising and forming pores (channels) in the outer mitochondrial membrane. Without pores, cytochrome C stays in the mitochondrial intermembrane space -> no apoptosome -> no caspase 9 activation.

LECTURE 15: HUMAN GENOME & GENE REGULATION

Q61. [PP] Fragile X syndrome: CGG repeat expansion in FMR1 gene disrupts which process?

- A. Splicing
- B. Translation
- C. Replication
- D. Transcription (silenced by hypermethylation)
- E. Post-translational modification

Answer: D

When CGG repeats >200, the FMR1 promoter and repeat region become hypermethylated -> heterochromatin formation -> transcription is SILENCED. No FMRP protein -> intellectual disability, autism-like behaviour, macro-orchidism. [PP: FA2 Med33-34 Q7, Re-exam Q1]

Q62. [PP] Features of Turner syndrome (45,XO) include all EXCEPT:

- A. Short stature
- B. Webbed neck
- C. Infertility
- D. Normal or near-normal intelligence
- E. Gynecomastia (breast development in males)

Answer: E

Turner syndrome (45,XO): webbed neck, low-set ears, low hairline, short stature, broad chest, no menstrual periods, infertility, normal intelligence. Gynecomastia is a feature of KLINEFELTER syndrome (47,XXY) not Turner's. Turner affects FEMALES. [PP: FA2 Sec11 Q9, Key Simulation Q1]

Q63. [PP] The Mediator complex in eukaryotic transcription functions as:

- A. RNA polymerase subunit
- B. Bridge connecting activators (at enhancers) to transcription factors (at promoter)
- C. Sigma factor equivalent
- D. Repressor protein
- E. Spliceosome component

Answer: B

Eukaryotic transcription initiation complex: activators bind distal enhancers -> DNA bending protein brings elements close -> Mediator bridges activator to TFs at core promoter -> RNA pol II recruited -> transcription initiates. [PP: Key Simulation Q5, Re-exam]

Q64. [PP] A promoter mutation most DIRECTLY alters which of the following?

- A. DNA polymerase activity
- B. RNA polymerase binding
- C. mRNA production
- D. Protein production
- E. Promoter site binding affinity

Answer: E

The promoter is the DNA sequence where RNA polymerase and transcription factors assemble. A mutation DIRECTLY changes the BINDING AFFINITY of these proteins to the promoter. This is the primary/direct effect. mRNA production (C) and protein (D) are downstream consequences. [PP: SA2 MED35 Q43, Re-exam Q11]

Q65. [PP] For MAXIMAL expression of the lac operon, what conditions must exist?

- A. High cAMP and low lactose
- B. Low glucose (CAP site occupied) AND lactose present (repressor inactivated)
- C. High glucose and high lactose
- D. Attenuation stem-loop formation
- E. Repressor bound to operator

Answer: B

Lac operon requires TWO conditions for maximal expression: (1) Lactose present -> allolactose binds repressor -> repressor leaves operator (derepression); (2) Glucose absent -> high cAMP -> cAMP-CAP binds CAP site -> activates transcription. Both needed simultaneously. [PP: Key Simulation Q6]

Q66. [PP] Estrogen receptor (ER) is classified as which type of receptor?

- A. Nuclear receptor (ligand-activated transcription factor)
- B. Cell membrane receptor
- C. Membrane enzyme (receptor tyrosine kinase)
- D. Adhesion molecule
- E. Ion channel receptor

Answer: A

Estrogen is lipid-soluble -> crosses plasma membrane -> binds cytoplasmic ER -> ER dimerises -> translocates to nucleus -> binds estrogen response elements (EREs) in DNA -> activates gene transcription. ER is a member of the nuclear receptor superfamily (steroid hormone receptors). [PP: SA2 MED35 Q12]

Q67. [PP] A riboswitch in bacterial mRNA regulates gene expression at the TRANSCRIPTIONAL level when:

- A. The aptamer binds its metabolite causing ribosome stalling
- B. The aptamer binds metabolite -> formation of terminator stem-loop + poly-U -> RNA pol falls off
- C. The aptamer unfolds mRNA for ribosome access
- D. The aptamer increases mRNA stability
- E. The aptamer recruits sigma factor

Answer: B

Transcriptional riboswitch: when metabolite (e.g., TPP, riboflavin) binds the aptamer domain, the expression platform forms a Rho-independent TERMINATOR stem-loop (+ run of U residues) -> RNA polymerase terminates prematurely -> gene OFF. If metabolite absent: antiterminator forms -> transcription continues -> gene ON. [PP: SA2 Med34 Q5 riboswitch]

Q68. [PP] Klinefelter syndrome (47,XXY) is caused by:

- A. Deletion of X chromosome
- B. Inversion of chromosome Y
- C. Non-disjunction during meiosis resulting in extra X chromosome in males
- D. Trinucleotide repeat expansion
- E. Robertsonian translocation

Answer: C

Klinefelter (47,XXY): non-disjunction during maternal meiosis I/II (or paternal) -> egg with 2 sex chromosomes -> fertilised by Y sperm -> 47,XXY. Features: tall, small testes, azoospermia (infertility), gynecomastia, normal intelligence. [PP: Lec15, SA2 Med36]

Q69. AGO2-mediated mRNA degradation (siRNA pathway) requires:

- A. Perfect complementarity between siRNA and target mRNA
- B. Imperfect complementarity (typical of miRNA)
- C. DNA methylation of target gene
- D. Histone deacetylation
- E. Nuclear localisation

Answer: A

siRNA pathway (AGO2, slicer activity): PERFECT complementarity between siRNA and mRNA -> AGO2 cleaves target mRNA at the centre of the duplex (mRNA degradation). miRNA pathway (AGO1-4): IMPERFECT complementarity -> translational repression or mRNA deadenylation. AGO2 is the only Argonaute with catalytic slicer activity. [PP: FA2 Sec11 Q8]

Q70. Chromosomal notation for Down syndrome (trisomy 21) in a male is:

- A. 45,XY
- B. 46,XY
- C. 47,XY,+21
- D. 47,XXY
- E. 46,XY,t(14;21)

Answer: C

Down syndrome: 47 chromosomes with an extra copy of chromosome 21 (trisomy 21). Male: 47,XY,+21. Female: 47,XX,+21. Most common form: non-disjunction (94%). Translocation Down syndrome: 46 chromosomes but extra copy of chr 21 attached to another chromosome (14) -> 46,XY,t(14;21). [PP: Lec15]

Q71. DNA methylation in gene regulation typically results in:

- A. Gene activation
- B. Histone acetylation
- C. Gene silencing (heterochromatin formation)
- D. Increased RNA polymerase binding
- E. Enhanced enhancer activity

Answer: C

CpG methylation (by DNA methyltransferases, DNMTs) at promoter regions recruits methyl-CpG binding proteins -> recruit histone deacetylases (HDACs) -> condensed chromatin (heterochromatin) -> gene SILENCING. TET enzymes remove methylation (demethylation) -> gene reactivation. This is epigenetic regulation. [PP: Lec15]

Q72. Which VNTR/STR feature makes it useful for DNA fingerprinting (forensic identification)?

- A. All humans have identical VNTR sequences
- B. VNTR regions code for essential proteins
- C. VNTR repeat numbers vary between individuals, making each person's pattern unique
- D. VNTRs are located only on sex chromosomes
- E. VNTRs cannot be detected by PCR

Answer: C

VNTRs (Variable Number Tandem Repeats) and STRs (Short Tandem Repeats) are repeated DNA sequences where the NUMBER of repeats varies between individuals (polymorphic). This variation creates unique band patterns ('fingerprints'). The probability of two unrelated people having identical patterns at multiple loci is ~ 1 in 10^{17} . Used in forensics, paternity testing, and CODIS database. [PP: Genetic Engineering Lec, Re-exam Q27]

LECTURE 16: BLOOD & HEME METABOLISM

Q73. [PP] What is the rate-limiting enzyme of heme synthesis?

- A. ALA dehydratase (ALAD)
- B. Delta-aminolevulinic acid synthase (ALAS)
- C. Porphobilinogen deaminase
- D. Ferrochelatase
- E. Uroporphyrinogen decarboxylase

Answer: B

ALAS catalyses the *FIRST* and rate-limiting step: glycine + succinyl-CoA → delta-ALA. Regulated by haem (feedback inhibition of ALAS). Requires pyridoxal phosphate (PLP, Vitamin B6) as cofactor. Located in the mitochondrial matrix. [PP: SA2 Med34 Q2, Re-exam Q6]

Q74. [PP] Which TCA cycle compound is the precursor for heme synthesis?

- A. Oxaloacetate
- B. Citrate
- C. Succinyl-CoA
- D. Fumarate
- E. Malate

Answer: C

Heme synthesis starts with succinyl-CoA (from TCA cycle) + glycine → ALA (catalysed by ALAS in mitochondria). This links energy metabolism to haem production. [PP: FA2 Sec11 Q13, FA2 Med33-34 Q13, Re-exam Q11]

Q75. [PP] Lead (Pb) poisoning inhibits which TWO enzymes in heme synthesis?

- A. ALAS and ferrochelatase
- B. ALA dehydratase (ALAD) and ferrochelatase
- C. Porphobilinogen deaminase and UROD
- D. ALAS and uroporphyrinogen synthase
- E. Ferrochelatase and porphobilinogen deaminase

Answer: B

Lead inhibits: (1) ALAD (2 ALA → porphobilinogen) → ALA accumulates in urine, neurotoxic; (2) Ferrochelatase (inserts Fe²⁺ into protoporphyrin IX → haem) → zinc protoporphyrin accumulates instead of haem. Result: microcytic anaemia + basophilic stippling + ALA in urine. [PP: SA2 Med34 Q3, BCH212 Re-exam Q65]

Q76. [PP] Porphyria Cutanea Tarda (PCT) is the most common porphyria, caused by deficiency of:

- A. ALAS
- B. ALAD
- C. Porphobilinogen deaminase
- D. Uroporphyrinogen decarboxylase (UROD)
- E. Ferrochelatase

Answer: D

PCT: UROD deficiency → uroporphyrinogens accumulate → PHOTSENSITIVITY (blistering skin in sun-exposed areas), hyperpigmentation, hypertrichosis. NO neurological symptoms (unlike AIP). Triggered by alcohol, HCV, iron, oestrogen. Most common porphyria. [PP: FA2 Sec11 Q14, Re-exam Q1]

Q77. [PP] Acute Intermittent Porphyria (AIP) is characterised by all EXCEPT:

- A. Acute abdominal pain
- B. Neuropsychiatric symptoms
- C. Port-wine coloured urine
- D. Photosensitivity and skin blistering
- E. Precipitated by barbiturates and certain drugs

Answer: D

AIP (porphobilinogen deaminase deficiency): ALA and PBG accumulate -> NEUROLOGICAL symptoms (severe abdominal pain, peripheral neuropathy, psychiatric symptoms, autonomic dysfunction, port-wine urine). NO PHOTOSENSITIVITY - porphyrins don't accumulate in skin. Barbiturates induce ALAS -> worsen attacks. [PP: Key Simulation Q, Re-exam Q8]

Q78. [PP] Which enzyme requires Vitamin B6 (PLP) as cofactor in heme synthesis?

- A. ALA synthase (ALAS)
- B. ALA dehydratase
- C. Porphobilinogen deaminase
- D. Uroporphyrinogen decarboxylase
- E. Ferrochelatase

Answer: A

ALAS: glycine + succinyl-CoA -> ALA + CO₂. This condensation reaction requires PLP (pyridoxal phosphate, from Vitamin B6) as cofactor - forms a Schiff base with glycine. B6 deficiency can cause sideroblastic anaemia due to impaired haem synthesis. [PP: FA2 Sec11 Q15, BCH212 Re-exam Q4]

Q79. [PP] Mature red blood cells (erythrocytes) cannot synthesise haem because they:

- A. Lack haem synthesis enzymes
- B. Have no mitochondria (ALAS is mitochondrial)
- C. Lack succinyl-CoA
- D. Have degraded all DNA
- E. Answers B and C are both correct

Answer: B

Mature RBCs have no nucleus AND no mitochondria. ALAS (first step) is located in the MITOCHONDRIAL MATRIX. Without mitochondria, the first step of haem synthesis cannot occur. Haem is synthesised during erythroblast development in bone marrow. [PP: SA2 Med34 Q1, Re-exam Q22]

Q80. Which amino acid, together with succinyl-CoA, is the starting material for heme synthesis?

- A. Alanine
- B. Serine
- C. Glycine
- D. Tyrosine
- E. Cysteine

Answer: C

ALAS condenses glycine (provides carbon 1 and nitrogen of ALA) + succinyl-CoA (4 carbons) -> ALA (5-carbon compound, delta-aminolevulinic acid). Two ALA molecules are then joined to form porphobilinogen (by ALAD).

LECTURE 17: PLASMA PROTEIN & BLOOD CLOTTING FACTOR

Q81. [PP] The MAJOR plasma protein and its primary function are:

- A. Prealbumin - vitamin A transport
- B. Albumin - maintains plasma oncotic pressure
- C. Globulin - primary immune function
- D. Fibrinogen - blood clotting
- E. AFP - growth factor

Answer: B

Albumin: ~60% of plasma protein, 3.5-5 g/dL, synthesised exclusively by liver (12g/day). Primary function: COLLOID OSMOTIC (oncotic) PRESSURE - keeps fluid in vessels. Also transports fatty acids, bilirubin, hormones, drugs. Half-life 15-20 days. [PP: FA2 Sec11 Q18]

Q82. [PP] What is the FIRST step in haemostasis after blood vessel injury?

- A. Fibrin clot formation
- B. Platelet aggregation
- C. Blood vessel constriction (vasoconstriction)
- D. Serotonin secretion
- E. Fibrinolysis

Answer: C

Haemostasis sequence: (1) VASOCONSTRICTION - local reflex reduces blood flow; (2) PRIMARY haemostasis - platelet adhesion (vWF-GPIb/IX-collagen) + aggregation (GPIIb/IIIa-fibrinogen) = platelet plug; (3) SECONDARY haemostasis - coagulation cascade -> fibrin; (4) Fibrinolysis - plasmin dissolves clot. [PP: FA2 Sec11 Q16]

Q83. [PP] Which enzyme digests fibrin clots during fibrinolysis?

- A. Fibrinogen
- B. Plasmin
- C. Dicoumarol
- D. Thrombin
- E. Protein C

Answer: B

Plasmin (serine protease) cleaves fibrin -> fibrin degradation products. Plasminogen converted to plasmin by t-PA (tissue plasminogen activator) and urokinase. Thrombin converts fibrinogen -> fibrin (opposite action). Alteplase and streptokinase are thrombolytic drugs that activate plasminogen. [PP: FA2 Sec11 Q17, Med36]

Q84. [PP] Alpha-1 antitrypsin (AAT) deficiency primarily causes which disease?

- A. Multiple myeloma
- B. Von Willebrand disease
- C. Emphysema (panacinar) and liver cirrhosis
- D. Haemophilia A
- E. DIC

Answer: C

AAT inhibits neutrophil elastase in lungs. Without AAT, elastase degrades lung elastin -> EMPHYSEMA (panacinar pattern, lower lobe predominant). Misfolded Z-type AAT also accumulates in hepatocytes -> cirrhosis. Smoking inhibits AAT-elastase binding -> worse disease. [PP: Med36 Q3, Lec16-17]

Q85. [PP] Multiple myeloma is associated with elevated levels of which plasma protein fraction?

- A. Albumin (elevated)
- B. Alpha-1 globulin
- C. Alpha-2 globulin
- D. Gamma globulin (immunoglobulins)
- E. Prealbumin

Answer: D

Multiple myeloma = monoclonal plasma cell proliferation producing abnormal immunoglobulins (M-protein, paraprotein). Serum electrophoresis shows a narrow M-spike in the gamma region. Gamma globulins = immunoglobulins (IgG, IgA, IgM, IgD, IgE), produced by plasma cells (B lymphocytes). [PP: Med36 Q1]

Q86. [PP] Aspirin prevents thrombosis by inhibiting:

- A. Fibrinolysis
- B. Prostaglandin and thromboxane synthesis via irreversible COX inhibition
- C. Vitamin K recycling
- D. Thrombin formation
- E. Antithrombin III

Answer: B

Aspirin irreversibly acetylates COX-1 (and COX-2) -> prevents synthesis of thromboxane A2 (TXA2) in platelets. TXA2 normally promotes platelet aggregation and vasoconstriction. Platelets have no nucleus -> cannot resynthesise COX -> effect lasts platelet lifetime (7-10 days). [PP: SA2 MED35 Q11, Key Simulation Q2]

Q87. [PP] Vitamin K deficiency leads to which coagulation abnormality?

- A. Increased thrombin production
- B. Prolonged prothrombin time (PT) due to deficiency of factors II,VII,IX,X
- C. Thrombocytopenia
- D. Elevated fibrinogen
- E. Decreased antithrombin III

Answer: B

Vitamin K is a cofactor for gamma-carboxylation of glutamate residues in clotting factors II (prothrombin), VII, IX, X (and anticoagulant proteins C and S). Without carboxylation, these factors cannot bind Ca²⁺ and are non-functional -> prolonged PT (extrinsic + common pathway). [PP: SA2 MED35 Q45, Re-exam Q23]

Q88. [PP] Haemophilia A is caused by deficiency of which clotting factor?

- A. Factor V
- B. Factor VII
- C. Factor VIII
- D. Factor IX
- E. Factor X

Answer: C

Haemophilia A (most common, 80% of haemophilia): Factor VIII deficiency. X-linked recessive (males affected). Prolonged aPTT (intrinsic pathway). Haemophilia B (Christmas disease): Factor IX deficiency, also X-linked. Both: haemarthrosis, easy bruising, prolonged bleeding. [PP: Lec16-17 notes, Re-exam Q21 context]

Q89. [PP] Von Willebrand factor (vWF) functions by:

- A. Activating the intrinsic coagulation pathway
- B. Bridging platelet GP Ib/IX to exposed subendothelial collagen (platelet adhesion)
- C. Converting fibrinogen to fibrin
- D. Activating protein C
- E. Inhibiting thrombin

Answer: B

vWF (multimeric glycoprotein): in primary haemostasis, bridges platelets (via GP Ib/IX receptor) to exposed collagen at injury site. vWF also carries and protects Factor VIII from premature degradation. Von Willebrand disease = most common inherited bleeding disorder. [PP: Lec16-17]

Q90. Haptoglobin's function in plasma is to:

- A. Transport iron (Fe³⁺) in blood
- B. Bind free haemoglobin released from haemolysed RBCs, preventing kidney damage
- C. Transport copper
- D. Inhibit elastase
- E. Activate complement

Answer: B

Haptoglobin (alpha2-globulin): binds free Hb from lysed RBCs -> Hb-Hp complex cleared by reticuloendothelial system. Prevents: (1) iron loss in urine; (2) kidney damage from free Hb. In haemolysis: haptoglobin is CONSUMED -> low serum haptoglobin is a marker of haemolysis. Acute phase protein. [PP: Lec16-17]

Q91. Albumin does NOT function to transport which of the following?

- A. Fatty acids
- B. Bilirubin
- C. Iron (Fe³⁺)
- D. Drugs (e.g., warfarin, NSAIDs)
- E. Calcium (ionised fraction)

Answer: C

Albumin transports: fatty acids, unconjugated bilirubin, drugs (warfarin, phenytoin, NSAIDs), steroid hormones, thyroid hormone, calcium (binds ~40% of total plasma calcium). Iron transport is primarily carried by TRANSFERRIN (beta-globulin). Not albumin. [PP: BCH212 Re-exam Q37, Lec16-17]

Q92. t-PA (tissue plasminogen activator) causes:

- A. Fibrin formation from fibrinogen
- B. Platelet aggregation via GPIIb/IIIa
- C. Conversion of plasminogen to plasmin (fibrinolysis)
- D. Activation of Factor X
- E. Inhibition of thrombin

Answer: C

t-PA is produced by endothelial cells and is a serine protease that cleaves plasminogen -> plasmin, which dissolves fibrin clots. t-PA is fibrin-bound, making it locally active at clot sites. Recombinant t-PA (alteplase, tenecteplase) is used as thrombolytic therapy in acute MI and ischaemic stroke. [PP: Med36 Q7]

LECTURE 18: NUTRITION

Q93. [PP] Pellagra (dermatitis, diarrhoea, dementia, death) is caused by deficiency of:

- A. Vitamin B1 (Thiamine)
- B. Vitamin B2 (Riboflavin)
- C. Vitamin B3 (Niacin)
- D. Vitamin B6 (Pyridoxine)
- E. Vitamin B12 (Cobalamin)

Answer: C

Pellagra = NIACIN (B3) deficiency. The 4 Ds: Dermatitis (photosensitive, sun-exposed areas), Diarrhoea, Dementia, Death. Niacin is part of NAD⁺/NADP⁺. Can be synthesised from tryptophan (60mg Trp → 1mg niacin, requiring B6). Isoniazid (TB drug) → B6 deficiency → secondary pellagra. [PP: SA2 Med34 Q1, BCH212 Re-exam Q36]

Q94. [PP] Prolonged consumption of RAW EGG WHITES causes deficiency of:

- A. Thiamine (B1)
- B. Riboflavin (B2)
- C. Niacin (B3)
- D. Biotin (B7)
- E. Folate (B9)

Answer: D

Raw egg whites contain AVIDIN, a glycoprotein that binds biotin (vitamin B7/H) with very high affinity, preventing gut absorption. Cooking denatures avidin. Biotin is a cofactor for carboxylase enzymes (pyruvate carboxylase, acetyl-CoA carboxylase). Deficiency: alopecia, dermatitis, neurological symptoms. [PP: SA2 Med34, SA2 Med36 Q4, FA2 Med33-34 Q23, Re-exam Q5]

Q95. [PP] Macrocytic anaemia + high homocysteine + NORMAL methylmalonic acid indicates:

- A. Vitamin B12 deficiency
- B. Folate (B9) deficiency
- C. Vitamin B6 deficiency
- D. Vitamin C deficiency
- E. Iron deficiency

Answer: B

Both B12 and folate deficiency cause macrocytic anaemia + elevated homocysteine. KEY differentiator: methylmalonic acid (MMA). B12 deficiency: MMA ELEVATED (methylmalonyl-CoA mutase needs B12). Folate deficiency: MMA NORMAL. B12 deficiency also causes neurological symptoms (subacute combined degeneration). [PP: SA2 Med36 Q3]

Q96. [PP] Vitamin D deficiency causes:

- A. Hypercalcaemia and hyperphosphataemia
- B. Hypocalcaemia and hypophosphataemia → rickets/osteomalacia
- C. Night blindness
- D. Megaloblastic anaemia
- E. Scurvy

Answer: B

VitD → increases intestinal Ca²⁺ and PO₄ absorption + bone mineralisation. Deficiency → low Ca²⁺, low PO₄ → secondary hyperparathyroidism (PTH rises) → Rickets (children, bowed legs, rachitic rosary) / Osteomalacia (adults, bone pain, pseudofractures). Hypercalcaemia = VitD TOXICITY. [PP: SA2 Med34 Lec18 Q3]

Q97. [PP] Vitamin C acts as cofactor for which enzyme in collagen synthesis?

- A. Carboxylase
- B. Prolyl/Lysyl Hydroxylase
- C. Decarboxylase
- D. Hydratase
- E. Kinase

Answer: B

Prolyl hydroxylase and lysyl hydroxylase require Vitamin C as electron donor to maintain iron (Fe²⁺, active form). These enzymes hydroxylate proline → hydroxyproline and lysine → hydroxylysine in collagen, essential for triple helix stabilisation and crosslinking. Deficiency → SCURVY: bleeding gums, perifollicular haemorrhages, poor wound healing. [PP: SA2 Med34 Q2, Re-exam Q12]

Q98. [PP] Cretinism in newborns is caused by severe deficiency of:

- A. Iron
- B. Iodine (→ thyroid hormone deficiency)
- C. Folate
- D. Vitamin B12
- E. Zinc

Answer: B

Iodine → thyroid hormone (T₃, T₄) synthesis. Severe iodine deficiency during pregnancy: maternal AND fetal hypothyroidism → cretinism: severe intellectual disability, stunted growth, deaf-mutism, spastic diplegia, goitre. In Thailand: common in mountainous regions (Chiang Rai, Nan) far from sea. [PP: SA2 MED35 Q71, SA2 Med34 Lec21 Q7, Re-exam Q2]

Q99. [PP] Neural tube defects (spina bifida, anencephaly) result from deficiency of:

- A. Folate (B9)
- B. Vitamin B12
- C. Vitamin D
- D. Vitamin A
- E. Biotin

Answer: A

Folate is critical for neural tube closure (days 21-28 of gestation - before most women know they are pregnant). Deficiency → failure of closure → spina bifida (open spine) or anencephaly (absent brain development). Recommendation: 400-800 mcg folate/day for all women of childbearing age. [PP: SA2 MED35, Re-exam Q10, Copy Q7]

Q100. [PP] Isoniazid (INH) for TB causes secondary deficiency of which vitamin?

- A. Vitamin B1
- B. Vitamin B3 (Niacin)
- C. Vitamin B6 (Pyridoxine)
- D. Vitamin B12
- E. Vitamin C

Answer: C

Isoniazid inhibits pyridoxal kinase → cannot activate Vitamin B6 to PLP → B6 deficiency → peripheral neuropathy, sideroblastic anaemia, secondary pellagra (B6 needed for niacin synthesis from tryptophan). Prevention: give pyridoxine (B6) supplements with INH. [PP: BCH212 Re-exam Q15, Copy Q23]

Q101. [PP] The ACTIVE form of Vitamin D produced in the KIDNEY is:

- A. Cholecalciferol (Vit D3)
- B. Ergocalciferol (Vit D2)
- C. 25-hydroxycalciferol (calcidiol)
- D. 1,25-dihydroxycalciferol (calcitriol)
- E. 24,25-dihydroxycalciferol

Answer: D

Vitamin D pathway: skin (UV) -> D3 -> liver (25-hydroxylase) -> 25-OH-D3 (calcidiol, STORAGE form, long half-life 2-3 weeks, measured clinically) -> kidney (1-alpha-hydroxylase) -> 1,25-(OH)₂-D3 (calcitriol, ACTIVE form, short half-life 4-6 hours). Calcitriol acts like steroid hormone. [PP: SA2 Med34 Lec21 Q9, Copy Q82]

Q102. [PP] Kwashiorkor features oedema but NOT Marasmus. This is because:

- A. Kwashiorkor has more weight loss
- B. Kwashiorkor has lower albumin (protein deficiency) -> low oncotic pressure -> fluid leaks to interstitium
- C. Marasmus has more severe calorie restriction
- D. Kwashiorkor affects younger children
- E. Marasmus affects liver more severely

Answer: B

Kwashiorkor = protein deficiency (often adequate calories): low albumin -> low oncotic pressure -> fluid leaks into interstitium -> OEDEMA (pitting oedema, ascites). Also: moon face, skin lesions, fatty liver, apathy. Marasmus = total calorie restriction: severe wasting/emaciation but NO oedema (albumin often preserved initially). [PP: SA2 Med34 Lec21 Q8, Med36 Q4, Re-exam Q3]

Q103. [PP] Prolonged PPI (proton pump inhibitor) use most commonly causes deficiency of:

- A. Vitamin B1
- B. Folate
- C. Vitamin B12
- D. Vitamin D
- E. Vitamin K

Answer: C

B12 requires gastric acid to release it from food proteins -> binds intrinsic factor (from parietal cells) -> absorbed in terminal ileum. PPIs suppress acid -> B12 remains bound to food protein -> cannot bind IF -> B12 malabsorption. Risk increases with >2 years of use. [PP: SA2 Med36 Q1, SA2 Med34 Lec21 Q4]

Q104. [PP] Which is TRUE about fat-soluble vitamins (A, D, E, K)?

- A. Mainly act as coenzymes like B vitamins
- B. Stored in body fat and liver; can accumulate to toxic levels
- C. Easily excreted in urine
- D. Require frequent daily intake like water-soluble vitamins
- E. Excessive intake is always safe

Answer: B

Fat-soluble vitamins: stored in adipose tissue and liver. Not water-soluble -> not easily excreted in urine -> accumulate with excess intake (hypervitaminosis). Most dangerous: Vit A (teratogen in pregnancy, liver damage) and Vit D (hypercalcaemia). Water-soluble vitamins (B-complex, C) act as coenzymes, excreted in urine. [PP: FA2 Sec11 Q19]

Q105. Vitamin K is required as a cofactor for carboxylation of which clotting factor residues?

- A. Serine residues in factor XIII
- B. Glutamate residues in factors II, VII, IX, X (gamma-carboxylation)
- C. Lysine residues in factor VIII
- D. Cysteine residues in factor V
- E. Threonine residues in fibrinogen

Answer: B

Vitamin K is essential cofactor for gamma-glutamyl carboxylase, which adds carboxyl groups to glutamate residues in factors II, VII, IX, X (and proteins C, S). Gamma-carboxyglutamate (Gla) residues bind Ca^{2+} , enabling factor-membrane interactions essential for coagulation. Warfarin/dicoumarol block VitK recycling. [PP: Lec16-17, BCH212 Re-exam Q70]

Q106. Vitamin C (ascorbic acid) acts as an antioxidant by:

- A. Inhibiting free radical chain reactions and regenerating active Vitamin E
- B. Decomposing hydrogen peroxide directly
- C. Chelating iron to prevent Fenton reaction
- D. Inhibiting xanthine oxidase
- E. Activating superoxide dismutase

Answer: A

Vitamin C: water-soluble antioxidant in plasma and cytoplasm. It donates electrons to free radicals (scavenging) and regenerates active Vitamin E (tocopherol) from the tocopheroxyl radical after Vit E neutralises a lipid radical. Vitamin C is itself then regenerated by glutathione. Forms a cascading antioxidant network.

Q107. Which vitamin deficiency causes PELLAGRA in a patient on isoniazid treatment for TB?

- A. Niacin directly
- B. Vitamin B6 (pyridoxine) -> secondary niacin deficiency
- C. Tryptophan deficiency
- D. Riboflavin
- E. Folate

Answer: B

INH inhibits B6 (pyridoxine) activation. PLP (active B6) is needed for the enzyme that converts tryptophan -> quinolinic acid -> niacin. Without B6: tryptophan-to-niacin pathway blocked -> secondary NIACIN (B3) deficiency -> pellagra. Direct cause: B6 deficiency; secondary effect: niacin deficiency.

Q108. A 4-year-old boy presents with bone pain, bleeding gums, perifollicular haemorrhages, and poor wound healing. Which vitamin is deficient?

- A. Vitamin B2
- B. Vitamin B9 (Folate)
- C. Vitamin C (Ascorbic acid)
- D. Vitamin D
- E. Vitamin K

Answer: C

Scurvy = Vitamin C deficiency. Symptoms: bleeding gums (pericoronitis), perifollicular haemorrhages (corkscrew hairs), poor wound healing, bone pain (periosteal haemorrhages), anaemia. Due to impaired collagen synthesis (need VitC for prolyl/lysyl hydroxylase). Rare in adults but can occur in children with restricted diets. [PP: FA2 Med33-34 Q24]

LECTURE 19: GENETIC ENGINEERING

Q109. [PP] Which technique is MOST appropriate for detecting SARS-CoV-2 viral LOAD quantitatively?

- A. ATK (Antigen Rapid Test)
- B. Conventional PCR
- C. RT-PCR
- D. RT-qPCR (reverse transcription quantitative PCR)
- E. ELISA

Answer: D

SARS-CoV-2 has RNA genome. RT-qPCR: (1) Reverse transcriptase: RNA → cDNA; (2) qPCR amplifies and QUANTIFIES in real time. Ct (cycle threshold) value inversely correlates with viral load. RT-PCR detects but doesn't quantify. ATK detects antigen (qualitative). ELISA detects antibodies. [PP: FA2 Sec11 Q22, Med36 Q1]

Q110. [PP] In forensic crime scene DNA analysis, which technique creates a DNA 'fingerprint' using VNTR probes?

- A. Northern blot
- B. Southern blot
- C. Western blot
- D. RT-PCR
- E. DNA microarray

Answer: B

Southern blot: DNA digested with restriction enzymes → gel electrophoresis → transferred to membrane → probed with VNTR probes → unique band pattern ('fingerprint') per individual. Modern forensics uses PCR-based STR analysis (CODIS database). [PP: SA2 MED35 Q29, SA2 Med34 Lec19 Q2, Re-exam Q27]

Q111. [PP] High-throughput gene expression profiling in cancer research uses:

- A. FISH
- B. RT-PCR
- C. PCR-RFLP
- D. DNA microarray
- E. Sanger sequencing

Answer: D

DNA microarray (gene chip): simultaneously measures thousands of genes' expression levels. mRNA extracted, reverse transcribed, labelled with fluorescent dye, hybridised to array → fluorescence indicates expressed genes. Used to identify cancer gene expression signatures. [PP: FA2 Sec11 Q23]

Q112. [PP] What method introduces foreign DNA into cells using an electrical field to create temporary membrane pores?

- A. Electrophoresis
- B. Electroporation
- C. Heat shock transformation
- D. Lipofection
- E. Viral transduction

Answer: B

Electroporation: brief high-voltage electric pulse creates temporary pores in the cell membrane. Negatively charged DNA moves through pores (toward anode). Pores reseal spontaneously. Used for cells resistant to other transformation methods. Heat shock uses Ca^{2+} + rapid heating to enhance DNA uptake. [PP: SA2 Med36 Q4, Re-exam Q14]

Q113. [PP] Why must RT-PCR be used to clone a eukaryotic gene for expression in bacteria?

- A. Bacteria grow faster
- B. Bacteria lack spliceosome machinery to remove introns; RT-PCR generates intron-free cDNA
- C. PCR cannot amplify eukaryotic DNA
- D. Bacteria require RNA as template
- E. Eukaryotic DNA is too large for PCR

Answer: B

Eukaryotic genes have introns. Bacteria CANNOT splice mRNA. RT-PCR uses reverse transcriptase to convert mRNA (already spliced, exons only) -> cDNA -> PCR amplification -> intron-free cDNA gene ready for bacterial expression. [PP: Lec19, Copy Q39, Re-exam Q15]

Q114. [PP] DNA ligase joins two DNA fragments by forming:

- A. Hydrogen bond
- B. Phosphodiester bond between 3'-OH and 5'-phosphate
- C. Peptide bond
- D. Glycosidic bond
- E. Van der Waals interaction

Answer: B

DNA ligase seals nicks by forming a phosphodiester bond between 3'-OH of one nucleotide and 5'-phosphate of the adjacent nucleotide, using NAD^+ (E.coli ligase) or ATP (T4 ligase) as energy. T4 DNA ligase joins both sticky and blunt ends. [PP: Key Simulation Lec19 Q2, Re-exam Q30]

Q115. [PP] The FASTEST method to screen bacterial colonies for correct gene insertion is:

- A. Replica plating
- B. Colony PCR
- C. RT-PCR
- D. Southern blotting
- E. Sanger sequencing after plasmid extraction

Answer: B

Colony PCR: directly pick cells from a colony -> add to PCR mix with insert-specific primers -> run PCR -> positive band on gel = correct clone. Takes 2-3 hours vs. days for extraction-based methods. [PP: Key Simulation Lec19 Q4, Re-exam Q32]

Q116. [PP] Western blotting detects specific proteins using:

- A. Fluorescent DNA probes
- B. VNTR probes
- C. Antibody specific to the target protein
- D. Complementary RNA probes
- E. Restriction enzymes

Answer: C

Western blot (immunoblotting): proteins separated by SDS-PAGE -> transferred to PVDF membrane -> blocked -> incubated with PRIMARY antibody (protein-specific) -> secondary antibody (enzyme/fluorochrome-conjugated) -> signal detection. Detects protein size and expression level. Northern: RNA; Southern: DNA. [PP: Key Simulation Lec19 Q5, Re-exam Q33]

Q117. [PP] Huntington's disease involves which mutation in the HTT gene?

- A. Missense point mutation (single aa change)
- B. Large exon deletion
- C. Excessive CAG trinucleotide repeat expansion (>36 repeats)
- D. Promoter methylation silencing HTT
- E. Nonsense mutation causing truncated protein

Answer: C

Huntington's disease: CAG repeat expansion (>36 repeats, normal: <36) in HTT exon 1 encodes polyglutamine (polyQ) tract in huntingtin protein. Toxic gain-of-function -> neurodegeneration (caudate, putamen). Autosomal dominant, anticipation (repeats expand in successive generations). CRISPR/Cas9 can silence mutant allele. [PP: SA2 MED35 on Huntington's, Lec19]

Q118. EcoRI restriction enzyme cuts 5'-G|AATTC-3'. What type of ends are produced?

- A. Blunt ends
- B. 5' overhangs (sticky ends)
- C. 3' overhangs (sticky ends)
- D. Nicks in only one strand
- E. Circular ends

Answer: B

EcoRI cuts: 5'-G|AATTC-3' and 3'-CTTAA|G-5' -> produces 5' overhangs (-AATT-): these are 5' STICKY ENDS that can anneal with complementary sticky ends from other EcoRI-cut fragments. 3' overhang examples: PstI (CTGCA|G). Blunt end cutters: SmaI (CCC|GGG), EcoRV.

Q119. PCR primer design guidelines include all EXCEPT:

- A. Primer length ~20 bases
- B. GC content 45-55%
- C. The 3' end should be G or C (GC clamp)
- D. Primers should form hairpins for specific binding
- E. Forward and reverse primers should have similar melting temperatures (T_m within 1°C)

Answer: D

Primer SHOULD NOT form hairpins or self-complementary structures (dimers). Hairpin formation prevents primer from annealing to template DNA. Correct guidelines: ~20 bp length; 45-55% GC content; G or C at 3' end; matching T_m (within 1°C); avoid repetitive regions; no self-complementarity or primer-dimer formation. [PP: Key Simulation Lec19 Q, Lec19]

Q120. Blue-white screening in bacterial transformation selects for:

- A. Antibiotic-sensitive colonies only
- B. Colonies with successfully INSERTED gene (white) vs intact lacZ (blue)
- C. Colonies expressing the most protein
- D. Largest colony size
- E. Colonies without any plasmid

Answer: B

Blue-white selection: The cloning site is WITHIN the lacZ gene on the plasmid. Intact lacZ -> beta-galactosidase -> cleaves X-gal -> BLUE colonies. Insert disrupts lacZ -> no functional enzyme -> WHITE colonies = successful insertion. All colonies must also be antibiotic-resistant (to verify plasmid uptake). [PP: Lec19, Re-exam Q13]

LECTURE 20: FREE RADICAL & ANTIOXIDANT

Q121. [PP] Which enzyme provides the FIRST LINE of enzymatic defence against superoxide (O₂⁻) radicals?

- A. Catalase
- B. Glutathione peroxidase (GPx)
- C. Superoxide dismutase (SOD)
- D. Glutathione reductase
- E. Peroxiredoxin

Answer: C

SOD converts: $2O_2^- + 2H^+ \rightarrow H_2O_2 + O_2$. Three forms: Cu/Zn-SOD (cytoplasm, 2 subunits), Mn-SOD (mitochondria), extracellular SOD. H₂O₂ is then detoxified by catalase (peroxisomes) or GPx (uses GSH). SOD is the PRIMARY enzymatic defence against superoxide. [PP: SA2 Med34 Lec20 Q1, FA2 Sec11 Q27, Re-exam Q3]

Q122. [PP] Which mineral is incorporated into Glutathione Peroxidase (GPx)?

- A. Iron (Fe)
- B. Manganese (Mn)
- C. Selenium (Se)
- D. Copper (Cu)
- E. Zinc (Zn)

Answer: C

GPx contains SELENOCYSTEINE (unusual amino acid with selenium in place of sulfur) at its active site. GPx uses glutathione (GSH) to reduce H₂O₂ and lipid hydroperoxides: $2GSH + H_2O_2 \rightarrow GSSG + 2H_2O$. Selenium deficiency $\rightarrow$ impaired GPx $\rightarrow$ increased oxidative damage. Mn: in Mn-SOD; Cu+Zn: in Cu/Zn-SOD; Fe: in catalase. [PP: FA2 Sec11 Q26, FA2 Med33-34 Q20, Re-exam Q2]

Q123. [PP] Which vitamin regenerates oxidised Vitamin E (tocopheroxyl radical) back to its active form?

- A. Vitamin A
- B. Vitamin C (Ascorbic acid)
- C. Vitamin D
- D. Vitamin K
- E. Vitamin B12

Answer: B

Vitamin E (alpha-tocopherol) donates an electron to neutralise a lipid peroxyl radical $\rightarrow$ tocopheroxyl radical (inactive). Vitamin C (water-soluble, in aqueous phase) donates an electron to regenerate active Vitamin E. Vitamin C is then regenerated by glutathione. This forms a network: $GSH \rightarrow Vit C \rightarrow Vit E \rightarrow$ membrane lipids. [PP: SA2 Med34 Lec20 Q3, Re-exam Q4]

Q124. [PP] A characteristic property of free radicals is all of the following EXCEPT:

- A. Extreme reactivity (unpaired electron)
- B. Small and diffusible
- C. Very short lifespan (microseconds)
- D. Long lifespan (hours to days)
- E. Generate new radicals by chain reaction

Answer: D

Free radical characteristics: unpaired electron -> extreme reactivity (A), small and diffusible (B), very SHORT lifespan (C, not long - highly reactive means they react immediately), cause chain reactions (E), damage tissues. Option D is INCORRECT - their lifespan is extremely SHORT (fractions of a second), not long. [PP: SA2 Med34 Lec20 Q2, Re-exam Q5]

Q125. [PP] Which part of amino acids is MOST susceptible to attack by free radicals?

- A. R-group (side chain)
- B. OH group
- C. Amino group (-NH₂)
- D. Carboxylic group (-COOH)
- E. Sulfhydryl group (-SH) in cysteine

Answer: E

Sulfhydryl groups (-SH) of cysteine are the most reactive targets for ROS oxidation -> disulfide bonds (S-S), mixed disulfide with glutathione (S-SG), or sulphenic acid. This can alter protein structure (crosslinking, aggregation) and inactivate enzymes with active site cysteines. Aromatic residues (Tyr, Trp, Phe) are also targets. [PP: FA2 Sec11 Q25, FA2 Med33-34 Q19, Re-exam Q1]

Q126. Cell MEMBRANE antioxidants include:

- A. Vitamin C and uric acid
- B. SOD and catalase
- C. Alpha-tocopherol (Vitamin E) and Coenzyme Q10
- D. Transferrin and ceruloplasmin
- E. Glutathione peroxidase and glutathione reductase

Answer: C

Classification by location: PLASMA: Vitamin C, uric acid, bilirubin, beta-carotene, albumin, ceruloplasmin, transferrin. CELL MEMBRANE: alpha-tocopherol (Vit E) and Coenzyme Q10 - both fat-soluble, embedded in membranes, protecting polyunsaturated lipids from peroxidation. INTRACELLULAR: SOD, catalase, GPx, glutathione. [PP: Free Radicals lecture]

Q127. Oxidative stress is defined as:

- A. The presence of any free radicals in the body
- B. Complete absence of antioxidants
- C. Imbalance between free radical production and antioxidant defence (excess ROS)
- D. DNA damage only
- E. Mitochondrial dysfunction only

Answer: C

Oxidative stress: imbalance where ROS/free radical PRODUCTION exceeds the cell's ANTIOXIDANT CAPACITY. Normally ROS production (metabolism, immune function) is balanced by antioxidants (enzymatic: SOD, catalase, GPx; non-enzymatic: Vit C, E, glutathione). Oxidative stress -> cellular damage -> contributes to ageing, cancer, cardiovascular disease, neurodegeneration. [PP: Lec20]

Q128. Superoxide anion (O_2^-) is primarily produced by which cellular structure?

- A. Nucleus
- B. Mitochondria (electron transport chain leakage)
- C. Endoplasmic reticulum
- D. Golgi apparatus
- E. Lysosomes

Answer: B

Under normal conditions, 3-5% of O_2 consumed by mitochondria is incompletely reduced to superoxide (O_2^-) due to electron 'leakage' from the electron transport chain (mainly at Complex I and Complex III). Mitochondria are the MAJOR endogenous source of ROS. Also produced by NADPH oxidase (phagocytes, respiratory burst), xanthine oxidase, CYP450 system. [PP: Lec20]

LECTURE 21: CLINICAL CORRELATION

Q129. [PP] Which oil has the HIGHEST proportion of saturated fatty acids?

- A. Corn oil
- B. Coconut oil
- C. Olive oil
- D. Canola oil
- E. Soybean oil

Answer: B

Coconut oil is ~87-92% saturated fatty acids (mainly lauric C12, myristic C14, caprylic C8). Olive oil: ~75% monounsaturated (oleic C18:1). Canola/soybean/corn: rich in polyunsaturated. High saturated fat -> raises LDL cholesterol -> atherosclerosis risk. [PP: SA2 Med34 Lec21 Q1, Re-exam Q1]

Q130. [PP] The antidote for paracetamol (acetaminophen) overdose is:

- A. EDTA (chelation)
- B. Deferoxamine (DFO)
- C. Naloxone
- D. Fomepizole
- E. N-acetylcysteine (NAC)

Answer: E

Paracetamol overdose: CYP2E1 produces toxic NAPQI -> depletes hepatic glutathione -> hepatocyte necrosis. NAC: (1) replenishes glutathione (cysteine donor); (2) directly inactivates NAPQI. Best within 8-10 hours. Naloxone: opioid antidote. Fomepizole: methanol/ethylene glycol antidote. DFO: iron overload chelator. [PP: SA2 Med36 Q2, SA2 Med34 Lec21 Q5, Re-exam Q8]

Q131. [PP] Which compound is secreted in BOTH bile and urine after phototherapy for neonatal jaundice?

- A. Urobilinogen
- B. Stercobilinogen
- C. Unconjugated bilirubin
- D. Lumirubin
- E. Urobilin

Answer: D

Phototherapy (blue light 425-475 nm) converts unconjugated bilirubin -> lumirubin (structural photoisomer, water-soluble WITHOUT needing conjugation). Lumirubin excreted directly in bile and urine. This bypasses the immature UGT1A1 enzyme in neonates. [PP: SA2 Med36 Q1, SA2 Med34 Lec21 Q3, Re-exam Q9]

Q132. [PP] Band keratopathy of the cornea is associated with:

- A. Obesity
- B. Type 1 Diabetes mellitus
- C. Hypocalcaemia
- D. Hypercalcaemia
- E. Hypothyroidism

Answer: D

Band keratopathy = calcium-phosphate crystal deposits in corneal Bowman's layer appearing as white/grey horizontal band. Associated with HYPERCALCAEMIA (hyperparathyroidism, Vit D toxicity, sarcoidosis, CKD with high phosphate). Also with chronic ocular inflammation (uveitis). [PP: SA2 Med36 Q3, Lec21 clinical case 11]

Q133. [PP] Which endocrine disrupting chemical (EDC) is found in polycarbonate plastics?

- A. Dioxins
- B. Lead
- C. Bisphenol A (BPA)
- D. Mercury
- E. Arsenic

Answer: C

BPA is a plasticiser used in polycarbonate plastics and epoxy resins (can linings). It leaches into food/drink and mimics oestrogen (xenoestrogen) -> endocrine disruption. Banned in children's products in many countries. Phthalates are also plastic plasticisers with EDC properties. [PP: SA2 Med34 Lec21 Q6, Re-exam Q4]

Q134. [PP] Which form of Vitamin D has the LONGEST half-life in circulation?

- A. 1,25-dihydroxycalciferol (calcitriol, active)
- B. 25-hydroxycalciferol (calcidiol, storage)
- C. Vitamin D2 (ergocalciferol)
- D. Vitamin D3 (cholecalciferol)
- E. 7-dehydrocholesterol (precursor)

Answer: B

25-OH-D3 (calcidiol) half-life: 2-3 WEEKS -> it is the STORAGE and clinically measured form. 1,25-(OH)₂-D3 (calcitriol, active) half-life: 4-6 HOURS (rapidly metabolised because it is the bioactive hormone). D3/D2 dietary forms: half-life hours (rapidly converted by liver). [PP: SA2 Med34 Lec21 Q9, Copy Q82]

Q135. [PP] Neural tube defects (spina bifida) are associated with deficiency of which nutrient?

- A. Iodine
- B. Iron
- C. Folate (Vitamin B9)
- D. Vitamin B12
- E. Zinc

Answer: C

Folate is essential for neural tube closure (days 21-28 after conception). Deficiency -> failure of closure -> spina bifida (myelomeningocele) or anencephaly. Encephalocele is also a NTD related to folate. Recommended: 400-800 mcg/day for all women of childbearing age. [PP: SA2 MED35, Re-exam Q10, Copy Q7]

Q136. [PP] Bitot's spots on the conjunctiva indicate deficiency of:

- A. Vitamin D
- B. Vitamin A
- C. Vitamin C
- D. Vitamin K
- E. Vitamin B12

Answer: B

Vitamin A deficiency progression in eyes: Night blindness (earliest, reversible) -> Conjunctival xerosis -> Bitot's spots (white, foamy, triangular keratinised plaques on conjunctiva) -> Corneal xerosis -> Keratomalacia (corneal softening/ulceration) -> Permanent blindness. [PP: Lec21 clinical, SA2 MED35 Q35, Copy Q61]

Q137. [PP] Phenylketonuria (PKU) - which amino acid is DECREASED in plasma?

- A. Alanine
- B. Cysteine
- C. Tyrosine
- D. Glutamine
- E. Phenylalanine (increased, not decreased)

Answer: C

PKU: phenylalanine hydroxylase (PAH) deficiency -> cannot convert Phe -> TYROSINE. Phe ACCUMULATES (neurotoxic). Tyrosine becomes CONDITIONALLY ESSENTIAL (depleted) -> DECREASED plasma tyrosine -> reduced dopamine, epinephrine, thyroxine, melanin. Treatment: low-Phe diet + tyrosine supplementation. [PP: SA2 MED35 Q16]

Q138. [PP] OGTT (Oral Glucose Tolerance Test) uses what dose of glucose?

- A. 25 g
- B. 50 g
- C. 75 g
- D. 100 g
- E. 150 g

Answer: C

Standard OGTT: fasting glucose measured -> drink 75g glucose in 250-300mL water -> measure glucose at 1 and 2 hours. Diagnostic: DM = 2-hr glucose $\geq$ 200 mg/dL; IGT = 140-199 mg/dL; NGT = <140 mg/dL. For gestational diabetes: sometimes 50g screening test or 100g diagnostic. [PP: SA2 MED35 Q80]

Q139. A 40-year-old alcoholic presents with wet beriberi (cardiac failure, oedema). The deficient vitamin is:

- A. Niacin (B3)
- B. Thiamine (B1)
- C. Riboflavin (B2)
- D. Vitamin B12
- E. Vitamin C

Answer: B

Wet beriberi = Thiamine (B1) deficiency + cardiovascular manifestations: high-output cardiac failure, cardiomegaly, oedema, tachycardia. Alcoholics: poor diet + impaired absorption. Thiamine -> TPP (cofactor for pyruvate dehydrogenase, alpha-KG DH, transketolase). Without TPP: pyruvate accumulates, ATP production impaired in heart. Dry beriberi: peripheral neuropathy. [PP: Lec21 clinical case 3]

Q140. Kayser-Fleischer rings and neuropsychiatric symptoms + liver disease suggest:

- A. Haemochromatosis
- B. Wilson's disease (copper accumulation, low ceruloplasmin)
- C. Alpha-1 antitrypsin deficiency
- D. Porphyria cutanea tarda
- E. Gilberts syndrome

Answer: B

Wilson's disease (ATP7B gene mutation, AR): copper accumulates in liver, brain, cornea, kidney. Kayser-Fleischer rings = copper in Descemet's membrane of cornea (brown/grey rings at corneal periphery). Ceruloplasmin markedly LOW. 24-hr urine copper HIGH. Treatment: penicillamine or zinc. [PP: Lec21 clinical case 17]

Q141. Acanthosis nigricans (dark velvety skin at neck and axillae) in an obese teenager most likely indicates:

- A. Vitamin A toxicity
- B. Insulin resistance and hyperinsulinaemia -> IGF-1 receptor activation in keratinocytes
- C. Addison's disease
- D. Primary hypothyroidism
- E. Cushing's syndrome

Answer: B

Benign acanthosis nigricans in young obese patients = insulin resistance -> hyperinsulinaemia -> insulin binds IGF-1 receptors on keratinocytes -> epidermal hyperplasia + hyperpigmentation. Associated with metabolic syndrome, type 2 DM risk. Malignant AN in older, thin patients: rapid onset suggests visceral malignancy (gastric cancer). [PP: Lec21 clinical case 12]

Q142. A patient with gallstones has elevated ALP and GGT but normal ALT and AST. This pattern suggests:

- A. Acute viral hepatitis (hepatocellular damage)
- B. Alcoholic hepatitis
- C. Cholestasis (biliary obstruction)
- D. Hepatocellular carcinoma
- E. Haemolytic anaemia

Answer: C

ALP and GGT are enzymes from biliary epithelial cells, elevated in CHOLESTASIS (bile duct obstruction). ALT/AST are HEPATOCELLULAR INJURY markers (liver cell damage). Normal ALT/AST + elevated ALP/GGT = cholestatic pattern. Gallstones blocking common bile duct -> post-hepatic (obstructive) jaundice. [PP: BCH212 Re-exam Q82, Copy Q51]

END OF SIMULATION EXAM

Question Summary

Lecture	Topic	Real Exam	This Sim	Past Paper Qs
Lec 10	DNA & RNA Synthesis	8 Qs	12 Qs	9 Qs
Lec 11	Protein Synthesis & Transport Protein	8 Qs	12 Qs	9 Qs
Lec 12	Liver Function	12 Qs	16 Qs	12 Qs
Lec 13	Thalassemia	6 Qs	10 Qs	8 Qs
Lec 14	Apoptosis	6 Qs	10 Qs	8 Qs
Lec 15	Human Genome & Gene Regulation	8 Qs	12 Qs	8 Qs
Lec 16	Blood & Heme Metabolism	4 Qs	8 Qs	7 Qs
Lec 17	Plasma Protein & Blood Clotting Factor	8 Qs	12 Qs	9 Qs
Lec 18	Nutrition	12 Qs	16 Qs	12 Qs
Lec 19	Genetic Engineering	8 Qs	12 Qs	9 Qs
Lec 20	Free Radical & Antioxidant	4 Qs	8 Qs	5 Qs
Lec 21	Clinical Correlation	10 Qs	14 Qs	10 Qs
TOTAL		94 Qs	142 Qs	106 Qs

Total: 142 questions | 106 from past papers [PP] | Real exam: 94 questions

Good luck on your SA2! Focus on [PP] questions first — they are the most likely to appear!